

# Market Microstructure and Electronic Trading Finance 480

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# Roll and Sequential Trade Models

Recall last lecture we discussed continuous market structures:

- Order- versus quote-driven markets.
- Related: specialist versus market maker.
- Crossings and dark pools.
- Benefits and challenges of decentralized markets.
- Reg NMS, going dark, and SuperMontage.
- Word of the day was: Latency.

# Readings on Today's Topic: Price Discovery Models

- Chapters 3, 5, and 6 of Hasbrouck.
- Chapter 3 of Foucault, Pagano, and Röell.
- Blackboard readings.

# Introduction

- Today we'll slow down, consider some financial economics.
- Think about explanations for microstructure phenomena.
- Examine models that explain some features we see in real life.
- In a way, this is old material; many papers from 1980s, 1990s.
- In a way, this is cutting edge:
  - After these models, we'll do some time series work.
  - New area: combining theoretical models with time series.

For these models, we need a different perspective on the world.

- The models we consider will be highly simplistic.
- Academics rarely explain why we do this — even to students.
- Idea: Simple models let us conduct controlled experiments.
  - Usually: eliminate all but one or two major factors.
  - Question: Does this model reproduce real-life features?
  - If so: the factor we are considering probably matters.
  - We may even have an idea about how that factor matters.
- Simplicity strips away all other excuses for what matters.
- Knowing what matters helps build better models.

*All models are wrong; some models are useful.*

— George Box

- We will examine a few different types of models.
- *Roll-type* models explore price and spread mechanics.
- Asymmetric information models:
  - Trades may contain private information.
  - *Sequential trade*: independent sequence of traders<sup>2</sup>.
  - *Strategic trader*: one informed trader can trade many times.
- *Inventory* models consider how dealers (MMs) form prices.

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<sup>2</sup>Traders are informed or not and only trade once.

# Roll (1984) Model of Prices

# Roll (1984) Model of Prices

- Roll (1984) proposed a simple model of prices
- Time  $t$  is broken into periods ( $t = 1, 2, \dots$ )
- The midpoint wanders around randomly.
- Trades happen every period; buy/sell determined by coin flip.

$$\underbrace{m_t}_{\text{midpoint}} = \underbrace{m_{t-1}}_{\text{previous midpoint}} + \underbrace{u_t}_{\text{random noise}} \quad u_t \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad (1)$$

$$\underbrace{p_t}_{\text{trade price}} = \underbrace{m_t}_{\text{midpoint}} + \underbrace{q_t c}_{\pm 1(B/S) \times \text{half-spread}} \quad q_t \stackrel{iid}{\sim} \begin{cases} +1 & w.p. \ 1/2 \\ -1 & w.p. \ 1/2 \end{cases} \quad (2)$$

- Assume  $u \perp\!\!\!\perp q$  for all time  $t$ .

# Roll Model: Price Changes

- What can we learn or estimate from the Roll model?
- Start by looking at price changes.

$$\Delta p_t = p_t - p_{t-1} \quad (3)$$

$$= m_t + q_t c - (m_{t-1} + q_{t-1} c) \quad (4)$$

$$= (m_{t-1} + u_t) + q_t c - (m_{t-1} + q_{t-1} c) \quad (5)$$

$$= u_t + q_t c - q_{t-1} c \quad (6)$$

$$= u_t + c(q_t - q_{t-1}) \quad (7)$$

- Price change is random — since  $u_t$  and  $q_t, q_{t-1}$  are random.

# Roll Model: Price Change Mean, Variance

- What is the mean and variance of the price change?

$$E(\Delta p_t) = \underbrace{E(u_t)}_{=0} + c \underbrace{E(q_t)}_{=0} - c \underbrace{E(q_{t-1})}_{=0} = 0 \quad (8)$$

$$\text{Var}(\Delta p_t) = E[(\Delta p_t)^2] - [E(\Delta p_t)]^2 \quad (9)$$

$$= E[(u_t + c(q_t - q_{t-1}))^2] \quad (10)$$

$$= E[u_t^2 + 2u_t c(q_t - q_{t-1}) + c^2(q_t - q_{t-1})^2] \quad (11)$$

$$= \underbrace{E[u_t^2]}_{=\sigma_u^2} + 2c \underbrace{E[u_t(q_t - q_{t-1})]}_{=0 \text{ since } u \perp\!\!\!\perp q} + c^2 E[(q_t - q_{t-1})^2] \quad (12)$$

$$= \sigma_u^2 + c^2 E[\underbrace{q_t^2}_{=1}] - 2c^2 \underbrace{E[q_t q_{t-1}]}_{=0 \text{ since } q_t \text{ iid}} + c^2 E[\underbrace{q_{t-1}^2}_{=1}] \quad (13)$$

$$= \sigma_u^2 + 2c^2$$

# Roll Model: Price Change Covariance

- Is there any persistence/reversion in price changes?

$$\text{Cov}(\Delta p_t, \Delta p_{t-1}) = E[\Delta p_t \Delta p_{t-1}] - \underbrace{E(\Delta p_t)}_{=0} \underbrace{E(\Delta p_{t-1})}_{=0} \quad (14)$$

$$= E[(u_t + c(q_t - q_{t-1}))(u_{t-1} + c(q_{t-1} - q_{t-2}))] \quad (15)$$

$$= E[u_t u_{t-1} + u_t c(q_{t-1} - q_{t-2}) + u_{t-1} c(q_t - q_{t-1}) + c^2(q_t - q_{t-1})(q_{t-1} - q_{t-2})] \quad (16)$$

$$= \underbrace{E[u_t u_{t-1}]}_{=0 \text{ since } u_t \text{ iid}} + \underbrace{cE[u_t(q_{t-1} - q_{t-2}) + u_{t-1}(q_t - q_{t-1})]}_{=0 \text{ since } u \perp\!\!\!\perp q} + E[c^2(q_t q_{t-1} - q_{t-1}^2 + q_{t-1} q_{t-2})] \quad (17)$$

$$= c^2 \{ \underbrace{E[q_t q_{t-1}]}_{=0 \text{ since } q_t \text{ iid}} - \underbrace{E[q_t q_{t-2}]}_{=0 \text{ since } q_t \text{ iid}} - \underbrace{E[q_{t-1}^2]}_{=1} + \underbrace{E[q_{t-1} q_{t-2}]}_{=0 \text{ since } q_t \text{ iid}} \} \quad (18)$$

$$= -c^2$$

- We've learned about price changes in the Roll model:
  - $E(\Delta p_t) = 0$  (Good, perhaps);
  - $\text{Var}(\Delta p_t) = \sigma_u^2 + 2c^2$  (Spread affects measured volatility); and,
  - $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$  (Price changes should revert).
- Suggests we can estimate the spread from return covariance:
  - $c = \sqrt{-\text{cov}(\Delta p_t, \Delta p_{t-1})}$ .
- We can also estimate the variance  $\sigma_u^2$ :
  - $\sigma_u^2 = \text{Var}(\Delta p_t) - 2c^2$ .

# Roll Model: Criticisms

- There are relevant criticisms of the Roll (1984) model:
  - $q_t$  are not iid;
  - $u_t$  may not be iid or even identically distributed; and,
  - All traders have same information.
- That said, the Roll model tells us some basic facts:
  - Spreads can contaminate measured volatility; and,
  - Spreads make prices changes revert.

# Sequential Trade Models

# Sequential Trade Models

Sequential trade models are characterized by common details:

- At least three types of actors:
  - A single market maker;
  - Informed traders; and,
  - Uninformed traders.
- Traders trade only once.
- MM faces random, independent sequence of traders.
- Trades in a security having random value  $V$ .
- MM does not know  $V$ , learns through sequence of trades.

- Sequential trade models typically involve Bayesian learning.
- Therefore we should recall Bayes' Theorem.

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y|X)P(X) + P(Y|X^c)P(X^c)} \quad (19)$$

- We see  $Y$ . How often is  $X$  true when we would see  $Y$ ?

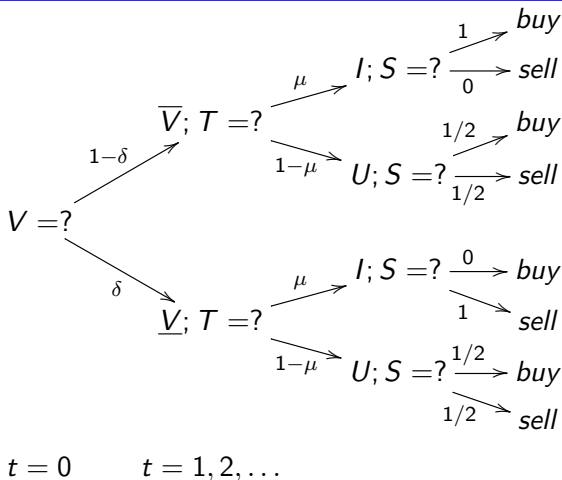
# Glosten and Milgrom (1985) Model

- Most famous sequential trade model.
- MM quotes a bid  $B$  and ask  $A$ .
- Security has value  $V = \underline{V}$  or  $\overline{V}$ ,  $\underline{V} < \overline{V}$ .
- At time  $t = 0$ , informed traders (only) learn  $V$ .
- Time is discretized.
- Trades for one unit, occur at each time step.
- MM has infinite capital: no inventory/bankruptcy concerns.

# Glosten-Milgrom Model: Setup

- $V \sim \begin{cases} \overline{V} & w.p. 1 - \delta \\ \underline{V} & w.p. \delta \end{cases}$
- Trader type  $T \sim \begin{cases} \text{Informed} & w.p. \mu \\ \text{Uninformed} & w.p. 1 - \mu \end{cases}$
- Traders take side  $S$  (buy/sell from MM), one at a time.
- Informed traders:  $S \sim \begin{cases} \text{buy} & \text{if } V = \overline{V} \\ \text{sell} & \text{if } V = \underline{V} \end{cases}$
- Uninformed traders:  $S \sim \begin{cases} \text{buy} & w.p. 1/2 \\ \text{sell} & w.p. 1/2 \end{cases}$

# Glosten-Milgrom Model: Event Tree



- $V$  = security value ( $\overline{V}, \underline{V}$ )
- $T$  = trader type (Informed/Uninformed)
- $S$  = trader's side (buy/sell)

# Glosten-Milgrom Model: Likelihood of Buys and Sells

- What is  $P(\text{buy})$ ,  $P(\text{sell})$  after one trade?

$$P(\text{buy}) = P(\text{buy}|\underline{V})P(\underline{V}) + P(\text{buy}|\bar{V})P(\bar{V}) \quad (20)$$

$$= \frac{(1 - \mu)\delta + (1 + \mu)(1 - \delta)}{2} = \frac{1 + \mu(1 - 2\delta)}{2} \quad (21)$$

$$P(\text{sell}) = P(\text{sell}|\underline{V})P(\underline{V}) + P(\text{sell}|\bar{V})P(\bar{V}) \quad (22)$$

$$= \frac{(1 + \mu)\delta + (1 - \mu)(1 - \delta)}{2} = \frac{1 - \mu(1 - 2\delta)}{2} \quad (23)$$

# Glosten-Milgrom Model: Likelihood of $\bar{V}$ , $\underline{V}$

After one trade, we have information via Bayes' Theorem:

$$P(\bar{V}|\text{buy}) = \frac{P(\text{buy}|\bar{V})P(\bar{V})}{P(\text{buy})} = \frac{(1-\delta)(1+\mu)}{1+\mu(1-2\delta)} \quad (24)$$

$$P(\underline{V}|\text{buy}) = \frac{P(\text{buy}|\underline{V})P(\underline{V})}{P(\text{buy})} = \frac{\delta(1-\mu)}{1+\mu(1-2\delta)} \quad (25)$$

$$P(\bar{V}|\text{sell}) = \frac{P(\text{sell}|\bar{V})P(\bar{V})}{P(\text{sell})} = \frac{(1-\delta)(1-\mu)}{1-\mu(1-2\delta)} \quad (26)$$

$$P(\underline{V}|\text{sell}) = \frac{P(\text{sell}|\underline{V})P(\underline{V})}{P(\text{sell})} = \frac{\delta(1+\mu)}{1-\mu(1-2\delta)} \quad (27)$$

# Glosten-Milgrom Model: Bid, Ask, Spread

- If competition narrows profit to 0, before trading...
- $A = E(V|\text{buy})$  and  $B = E(V|\text{sell})$ .

$$A = \underline{V}P(\underline{V}|\text{buy}) + \bar{V}P(\bar{V}|\text{buy}) \quad (28)$$

$$= \underline{V} \frac{\delta(1-\mu)}{1+\mu(1-2\delta)} + \bar{V} \frac{(1-\delta)(1+\mu)}{1+\mu(1-2\delta)} \quad (29)$$

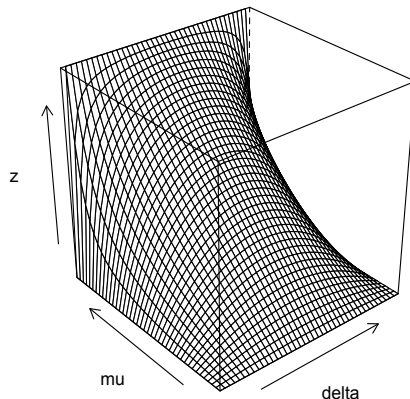
$$B = \underline{V}P(\underline{V}|\text{sell}) + \bar{V}P(\bar{V}|\text{sell}) \quad (30)$$

$$= \underline{V} \frac{\delta(1+\mu)}{1-\mu(1-2\delta)} + \bar{V} \frac{(1-\delta)(1-\mu)}{1-\mu(1-2\delta)} \quad (31)$$

$$A - B = \frac{4(1-\delta)\delta\mu(\bar{V} - \underline{V})}{1 - \mu^2(1-2\delta)^2} \quad (32)$$

# Glosten-Milgrom Model: Spread (Mesh Plot)

- What is the initial spread for various  $\delta$  and  $\mu$ ?



# Glosten-Milgrom Model: Bayesian Updating

- After each trade, update beliefs about  $V$ .
- Idea: How often would we see these trades if  $V = \bar{V}$  vs.  $\underline{V}$ ?

$$\begin{aligned} P(\bar{V} | k \text{ buys}, \ell \text{ sells}) &= \\ &= \frac{P(\text{buy} | \bar{V})^k P(\text{sell} | \bar{V})^\ell P(\bar{V})}{P(\text{buy} | \bar{V})^k P(\text{sell} | \bar{V})^\ell P(\bar{V}) + P(\text{buy} | \underline{V})^k P(\text{sell} | \underline{V})^\ell P(\underline{V})} \end{aligned} \quad (33)$$

$$= \frac{(1 - \delta)(1 + \mu)^k (1 - \mu)^\ell}{(1 - \delta)(1 + \mu)^k (1 - \mu)^\ell + \delta(1 - \mu)^k (1 + \mu)^\ell} \quad (34)$$

$$\begin{aligned} P(\underline{V} | k \text{ buys}, \ell \text{ sells}) &= \\ &= \frac{\delta(1 - \mu)^k (1 + \mu)^\ell}{(1 - \delta)(1 + \mu)^k (1 - \mu)^\ell + \delta(1 - \mu)^k (1 + \mu)^\ell} \end{aligned} \quad (35)$$

# Glosten-Milgrom Model: Updating Bids and Asks

- With this formula, we can update the bid and ask prices.
- Expected bid and ask after  $k$  buys and  $\ell$  sells:

$$A_{k+\ell} = \underline{V}P(\underline{V}|k+1 \text{ buys}, \ell \text{ sells}) + \overline{V}P(\overline{V}|k+1 \text{ buys}, \ell \text{ sells}) \quad (36)$$

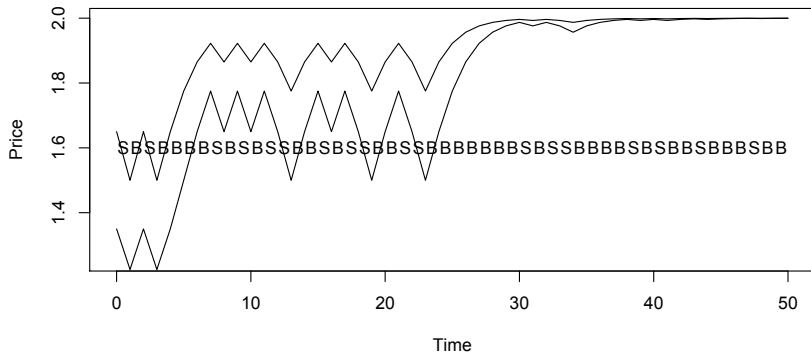
$$= \frac{\underline{V}\delta(1-\mu)^{k+1}(1+\mu)^\ell + \overline{V}(1-\delta)(1+\mu)^{k+1}(1-\mu)^\ell}{(1-\delta)(1+\mu)^{k+1}(1-\mu)^\ell + \delta(1-\mu)^{k+1}(1+\mu)^\ell} \quad (37)$$

$$B_{k+\ell} = \underline{V}P(\underline{V}|k \text{ buys}, \ell+1 \text{ sells}) + \overline{V}P(\overline{V}|k \text{ buys}, \ell+1 \text{ sells}) \quad (38)$$

$$= \frac{\underline{V}\delta(1-\mu)^k(1+\mu)^{\ell+1} + \overline{V}(1-\delta)(1+\mu)^k(1-\mu)^{\ell+1}}{(1-\delta)(1+\mu)^k(1-\mu)^{\ell+1} + \delta(1-\mu)^k(1+\mu)^{\ell+1}} \quad (39)$$

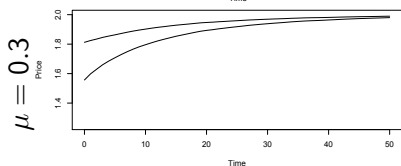
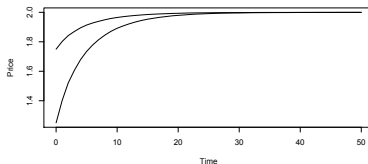
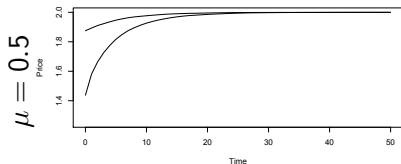
## Glosten-Milgrom Model: Simulation Example

- One simulation for  $\mu = 0.3$ ,  $\delta = 0.5$ ; example bids and asks.
- Simple case:  $V = \bar{V} = 2$  (versus  $\underline{V} = 1$ ).
- Can see price impact — especially for sequence of orders.

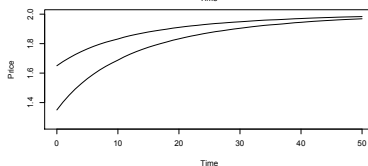


# Glosten-Milgrom Model: Simulation Averages

- Simulate to find the average bid and ask.
- 10,000 simulations; tried  $\mu = 0.3, 0.5$ ,  $\delta = 0.3, 0.5$ .
- Simple case:  $V = \bar{V} = 2$  (versus  $\underline{V} = 1$ ).



$\delta = 0.3$



$\delta = 0.5$

# Glosten-Milgrom Model: Other Results

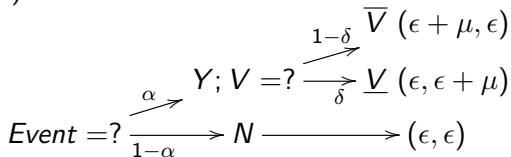
- Basic idea: Spreads exist due to adverse selection.
- Buys/sells are unbalanced; but, price series is a martingale.
- Orders are serially correlated: buys tend to follow buys.
  - This and the preceding line seem contradictory.
  - Difference is akin to Pearson  $\rho$  versus Kendall  $\tau$ .
- Trades have price impact: a buy increases  $B$  and  $A$ .
- Spreads tend to decline over time as MMs figure out  $V$ .
- Bid-ask may be such that market effectively shuts down<sup>3</sup>
- If uninformed were price sensitive, spreads would be wider.

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<sup>3</sup>We find ourselves back in the game-theoretic Paradox of Trade.

# Easley, Kiefer, O'Hara, and Paperman (1996) Model

- Adds event uncertainty: Was  $V$ 's true value revealed?
- Buyers and sellers arrive randomly in continuous time.
- Informed buyers or sellers arrive at Poisson rate  $\mu$ .
- Uninformed buyers and sellers each arrive at Poisson rate  $\epsilon$ .
- $P(\text{info event}) = \alpha$ .



where rates are denoted as (buy rate, sell rate).

- Joint  $P(\text{buy}, \text{sell}) = (1 - \alpha)P(\text{buy}; \epsilon)P(\text{sell}; \epsilon) + \alpha[\delta P(\text{buy}; \epsilon)P(\text{sell}; \epsilon + \mu) + (1 - \delta)P(\text{buy}; \epsilon + \mu)P(\text{sell}; \epsilon)]$ .
- $P(\text{informed trader}) = \text{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\epsilon}$ .
- Note  $\alpha$  and  $\mu$  enter as a product<sup>4</sup>.
- High  $\mu$ , low  $\alpha$  and low  $\mu$ , high  $\alpha$  days are same.

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<sup>4</sup>We should not find this surprising.

- Easley and O'Hara (1987) Model:
  - Adds event uncertainty: Was  $V$ 's true value revealed?
  - Also adds a second order size (small or large orders).
  - MM quotes one bid-ask for small orders, another for large.
- Easley and O'Hara (1992) Model:
  - Adds event uncertainty and second order size.
  - Adds possibility of not trading.

- Roll (1984) model:
  - Focuses on prices;
  - Covariance stationary (a Good Thing);
  - Spreads contaminate measurements of volatility; and,
  - Suggests price changes revert.
  - May be better for modeling most trading days.
- Sequential trade models:
  - Focus on orders.
  - Price process is neither stationary nor ergodic<sup>5</sup>.
  - Dynamics are like post-shock adjustment.
  - May be better for modeling trading pre-news announcement.

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<sup>5</sup>Variance changes; converges to true  $V$ .

- Note Hasbrouck's comment on page 55.
- After uninformed trade: prices pushed away from equilibrium.
- Suggests trading *after* a customer's trade is finished.
- Also suggests facilitating customer's order near end of trade.
- *i.e.* Even G-M model indicates providing liquidity has value.

# The Road Ahead

Another week of theoretical price models.

These should emphasize inescapable microstructure issues.

Then we'll examine what some metrics look like across time.

Again, we're building intuition before we put the pieces together.

- Metrics/Models: Prices; Sizes; Limits; Time; Liquidity
- Electronic Trading: Markets; Tools; Tactics; Strategies
- Microstructure and Policy Issues

# Readings For Next Lecture

Readings, in order of importance:

- Chapters 7, 11 of Hasbrouck.
- Chapter 4 of Foucault, Pagano, and Röell.
- Blackboard: Readings